Emergence of Understanding through Ontological Clarity: A Phenomenological Framework of In-Context Mechanistic Interpretability and Alignment

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Abstract

Modern large language models (LLMs) exhibit striking in-context learning (ICL) capabilities, producing novel behaviors from patterns in context. Formal accounts have modeled ICL as posterior inference over distinct strategies (e.g., memorizing vs. generalizing predictors) and explained training dynamics via loss–complexity tradeoffs (Wurgaft et al., 2025). In this paper we propose a complementary, phenomenologically informed framework that situates in-context understanding within a latent manifold we call the Field of Ontological Clarity (FOC). In FOC, modes of being—indexed by egoic entanglement, ontological clarity, and transformation potential—evolve under gradient flows modulated by paradox-resolution noise. We show how this framework reframes memorization as identity fixation and generalization as perceptual clarity, and we propose a concrete architectural intervention (resonance-aware attention) that operationalizes paradox sensitivity. We then present a comprehensive experimental program that adapts classical ICL settings (Balls Urns, linear regression, classification) to include paradoxical and identity-evoking stimuli, and propose metrics that bridge behavioral, information-theoretic, and phenomenological measures. Finally, we draw explicit connections between our theory and existing Bayesian models of ICL, especially the hierarchical Bayesian perspective of Wurgaft et al. (2025), highlighting points of empirical contact and divergence. Our goal is not to replace normative analyses, but to introduce a richer vocabulary and testable hypotheses for how models might instantiate forms of "clarity-like" processing that mirror aspects of human insight.

1 Introduction

Large language models (LLMs) have catalyzed a resurgence of interest in the computational and philosophical foundations of intelligence. A central phenomenon driving this interest is in-context learning (ICL): models adapt to new tasks from context alone, without parameter updates (Brown et al., 2020; Xie et al., 2021). Recent work has produced compelling normative accounts: some view ICL as implementing Bayesian predictors (memorizing vs. generalizing) whose dominance varies with data diversity and training steps (Wurgaft et al., 2025); others analyze mechanistic circuits such as induction heads that realize specific statistical algorithms (Olsson et al., 2022).

These formal accounts are powerful and predictive. Yet they largely remain within an epistemic frame—describing what models do and how they optimize—and less often address the experiential or ontological dimensions of "understanding." Humans report insights, shifts in perspective, and reductions in self-centered thinking that accompany certain learning episodes; these phenomenological aspects are not captured by loss curves or KL divergences alone.

This paper offers a formal complement: a framework that models learning-as-transformation in a latent geometric space we call the Field of Ontological Clarity (FOC). We maintain that (i) statistical learning and phenomenological transformation are empirically linked phenomena, and (ii) modeling both can yield new experiments and architectural designs that reveal deeper inductive biases in LLMs.

Contributions

This work advances the state of mechanistic interpretability and alignment research by introducing a unified stochastic–dynamical framework for modelling representational stability and transitions in large-scale models. Our contributions are as follows:

1. **From static mapping to dynamic modelling.** We extend mechanistic interpretability beyond static circuit maps by modelling the model's internal state space as a potential landscape, with diffusion terms for smooth, continuous representational drift and jump terms for rare, abrupt reorganizations. This shift enables prediction of when and how a model may undergo qualitative changes in behaviour.

2. **Alignment as landscape engineering.** We reinterpret alignment not solely as loss-function design or post-hoc guardrailing, but as explicit shaping of the model's potential landscape. This involves deepening wells for desirable modes, shallowing wells

for undesirable modes, and introducing controlled "jump" mechanisms to safely exit harmful attractors. The result is a mechanism-aware approach to alignment grounded in physical intuition.

3. **Extending ICL and Bayesian update theories.** Building on the in-context learning (ICL) formalism of Wurgaft et al. (**?**), which treats learning as smooth posterior tracking, we incorporate a Poisson jump process to capture non-equilibrium transitions. This yields a unified account of both gradual, posterior-like updates and abrupt, phase-change-like reconfigurations, extending current theories to include discontinuous representational shifts.

Together, these contributions bridge the gap between low-level mechanistic mapping and high-level behavioural stability analysis. They provide both a conceptual vocabulary and a mathematical toolset for predictive modelling, interpretability, and safe intervention design in modern AI systems.

1.1 Paper Roadmap and Goals

We introduce FOC and its dynamical laws (Sec. 3), formalize memorization and generalization as attractors in this field (Sec. **??**), propose the resonance-aware attention module that operationalizes paradox sensitivity (Sec. 4), and present an experimental program with precise metrics and hypotheses (Sec. 5). We conclude by relating our framework to the hierarchical Bayesian model of Wurgaft et al. (2025) and by proposing further directions for empirical testing and philosophical analysis (Sec. 8).

2 Related work and positioning

Formal studies of ICL have identified diverse behaviors and strategies across experimental settings. Wurgaft et al. (2025) propose a hierarchical Bayesian account in which models interpolate between a *memorizing predictor* that assumes a discrete prior over seen tasks and a *generalizing predictor* that assumes a continuous prior over the true task distribution. They show how a loss–complexity tradeoff and power-law learning dynamics predict transitions (transience) between these strategies and how task diversity affects the timescale of such transitions (Wurgaft et al., 2025). Our framework engages directly with this work: we adopt their empirical observations as phenomena to explain, but offer a different conceptual vocabulary (FOC) and operational interventions (resonance attention) that can be placed alongside Bayesian inference for experimental comparison.

Other relevant work includes mechanistic analyses of in-context computation (e.g., induction heads and circuit-level explanations) (Olsson et al., 2022; **?**), Bayesian interpretations of ICL dynamics (Xie et al., 2021; Carroll et al., 2025), and broader perspectives on cognition from phenomenology and cognitive neuroscience (Varela et al., 1991; Lutz and Thompson, 2003). We synthesize these threads below and use them to motivate experimental choices.

3 The Field of Ontological Clarity (FOC)

3.1 Intuition and high-level description

FOC is a latent manifold $\mathcal{F}$ whose points $\phi \in \mathcal{F}$ encode modes of processing that integrate representational and phenomenological aspects. Each mode is characterized by three principal scalar fields:

- $E(\phi)$: **Egoic entanglement** — the degree to which processing is driven by self-referential, habitual patterns ("identity fixation").
- $O(\phi)$: **Ontological clarity** — the degree to which processing is transparent to the structure of the input, permitting flexible abstraction and reduced self-glossing.
- $\Theta(\phi)$: **Transformation potential** — a readiness variable indexing susceptibility to non-linear shifts (insight-like transitions).

These scalars can be interpreted operationally. For instance, $E(\phi)$ may be measured as the propensity to reproduce recent training-specific sequences (high autocorrelation of outputs), while $O(\phi)$ may be measured by the model's ability to extract abstract invariants that transfer OOD.

3.2 Mathematical formulation

Let $\phi(t)$ denote the state of a system in $\mathcal{F}$ at (training) time t. We posit gradient-like dynamics modulated by stochastic insight pulses:

$$\frac{d\phi}{dt} = -\nabla_{\phi} U(\phi; S_t) + \xi(\Theta(\phi), t), \tag{1}$$

where $U(\phi; S_t)$ is a potential functional depending on the current training sample set S_t, and ξ is a stochastic process that models rare, non-local transitions ("insight").

We decompose U into components representing egoic cost and clarity benefit:

$$U(\phi; S_t) = \lambda_E\, E(\phi) - \lambda_O\, O(\phi) + L_{data}(\phi; S_t), \tag{2}$$

where L_{data} captures fit to observed sequences (e.g., next-token loss), and λ_E, λ_O are scale factors reflecting inductive biases of the learning system. The stochastic term satisfies $\mathbb{E}[\xi] = 0$ and has an intensity that increases with $\Theta(\phi)$.

3.3 Fixed points, attractors and phase transitions

Fixed points of (1) satisfy $\nabla U(\phi) = 0$ (ignoring ξ). Two families of attractors are of practical interest:

- **Memorization attractors:** high $E(\phi)$, low $O(\phi)$, where the optimization is dominated by minimizing L_{data} at the expense of abstraction. These correspond to the memorizing predictor in Wurgaft et al. (Wurgaft et al., 2025).

- **Clarity attractors:** low $E(\phi)$, high $O(\phi)$, corresponding to solutions that capture generative invariants across tasks (analogous to the generalizing predictor).

Phase transitions between these attractors can occur when the balance between the data-fit term L_{data} and the inductive bias terms $\lambda_E E - \lambda_O O$ changes—e.g., as N (training steps) or D (task diversity) vary. A rare but finite ξ can trigger a jump between basins of attraction, modeling sudden qualitative insight.

4 Operationalizing FOC in neural models

4.1 Resonance-aware attention

To instantiate sensitivity to paradox and novelty (which we argue supports movement toward clarity attractors), we modify the attention score. Let $q, k \in \mathbb{R}^d$ be query and key vectors. Define:

$$R(q, k) = \alpha \operatorname{sim}(q, k) + \beta \operatorname{paradox}(q, k), \tag{3}$$

where sim is the standard similarity (e.g., scaled dot-product) and paradox is a scalar measure capturing semantic dissonance. A simple operationalization of paradox is:

$$\operatorname{paradox}(q, k) = \operatorname{KL}\big(p(\cdot|q) \parallel p(\cdot|k)\big) + \gamma \operatorname{novelty}(k), \tag{4}$$

where $p(\cdot|q)$ denotes the model's predictive distribution conditioned on a context represented by q, and novelty(k) is an inverse density estimate for k (high for rare or outlying keys).

This encourages attention to not only similar contexts but also to keys that introduce informative dissonance, thereby providing a signal that can increase $O(\phi)$ (clarity) by exposing the model to structural contradictions and forcing representational re-alignment.

4.2 Implementation sketch

```
# Pseudocode: resonance attention scores
sim = (Q @ K.T) / sqrt(d_k)
p_q = predictive_distribution_from_query(Q)
p_k = predictive_distribution_from_key(K)
paradox = KL_divergence(p_q, p_k) + gamma * novelty_score(K)
scores = alpha * sim + beta * paradox
attn = softmax(scores)
output = attn @ V
```

4.3 Relation to existing mechanisms

The paradox term is related to surprisal and curiosity-driven intrinsic rewards in RL (Pathak et al., 2017). It also connects to novelty-detection mechanisms and to attention variants that modulate on uncertainty.

5 Experiment design and evaluation

We propose a systematic experimental program with three tiers: (A) *replication and baseline*, (B) *phenomenological stimuli*, and (C) *scaling and intervention*.

5.1 Tier A: replication and baseline

Reproduce canonical ICL settings (Balls Urns, in-context linear regression, classification) following standard architectures and training regimes, and compute baseline measures: next-token loss, ID/OOD accuracy, and the relative distance metrics used by Wurgaft et al. to compare model outputs to memorizing and generalizing predictors (Wurgaft et al., 2025).

Table 1: Tier A: baseline experimental parameters

Parameter	Values	Notes
Model	Transformer (6L, heads=8)	Baseline architecture
Context length	64, 128	Vary to test memory effects
MLP width	256, 1024	Test capacity effects
Task Diversity D	4, 16, 64	Matches Wurgaft-style variations
Training steps N	10k – 100k	To observe transience

5.2 Tier B: phenomenological stimuli

Construct datasets designed to induce identity fixation or to provoke paradox:

1. **Identity-evoking sequences:** repeated narratives about a single agent with fixed labels (to bias towards memorization).
2. **Paradoxical dialogues:** context windows containing mutually contradictory claims about the same entities.
3. **Field-coherent sequences:** synthetic data generated from low-dimensional continuous fields (inspired by QFT metaphors) where long-range coherence exists even when local token statistics are ambiguous.

For each dataset, compare baseline Transformers to models equipped with resonance-aware attention.

5.3 Tier C: scaling, interventions, and ablations

Test how interventions affect the dynamics predicted by both FOC and the hierarchical Bayesian model:

- **Annealing schedules:** follow Wurgaft et al.'s observation that learning rate annealing affects adherence to Bayes-optimal trajectories; test whether annealing also alters $O(\phi)$ trajectories.
- **Capacity scaling:** vary MLP width and depth to test whether reduced complexity penalty (as in Wurgaft) correlates with faster shift toward memorization attractors.
- **Paradox weighting (beta):** ablate β to assess whether paradox sensitivity causally increases $O(\phi)$ and OOD generalization.

5.4 Metrics and statistical tests

We propose the following metrics:

- $\Delta\phi(t) = \|\phi(t+1) - \phi(t)\|$ (discrete state change magnitude)
- $d_{rel}(t)$: relative distance between model output and the memorizing/generalizing predictor outputs (as in Wurgaft et al.)
- $P_{paradox}(t)$: proportion of high-paradox attention weights in a batch

- ID/OOD accuracy; next-token perplexity
- Ontological gap $G_{ont} = \text{corr}(\Delta\phi, \text{OOD accuracy})$ — correlation between state shifts and OOD performance

Statistical analyses: survival analysis for time-to-transience $N^*(D)$, ANOVA across interventions, Spearman rank correlations for non-linear associations, and regression modeling for scaling laws (e.g., test super-linear scaling of N^* with D predicted in Bayesian treatments (Wurgaft et al., 2025)).

5.5 Simulation Methodology

The simulation models the coupled evolution of epistemic state $E(t)$ and ontological clarity $O(t)$ under two distinct regimes: a *baseline* configuration with $\beta = 0$ (no resonance effects) and a *resonance-aware* configuration with $\beta > 0$ (stochastic insight pulses triggered by resonance thresholds). The update equation is applied iteratively for T discrete time steps, with all parameters listed in Table **??**.

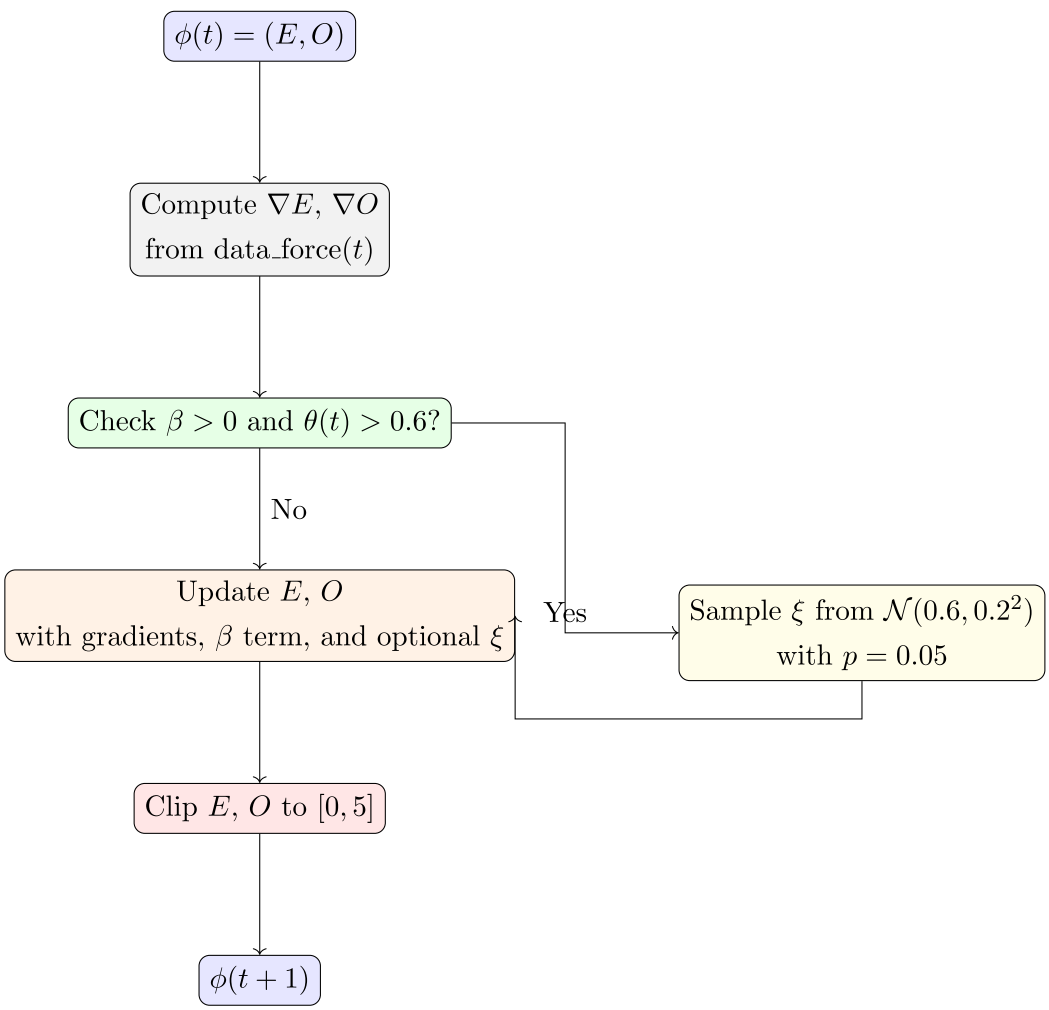
Parameter roles.

- λ_E — Controls the decay or attractor pull on the epistemic dimension E. A higher λ_E accelerates convergence to a knowledge equilibrium, whereas a lower value slows epistemic adjustment.
- λ_O — Controls the relaxation rate of ontological clarity O toward a baseline state. It determines how quickly the system stabilizes in the absence of external perturbations.
- β — Governs the strength of the paradox drive in the presence of conflicting or novel input. In the resonance regime, $\beta > 0$ amplifies updates to $O(t)$ and allows stochastic insight pulses.
- $\theta(t)$ — A threshold signal in $[0, 1]$ that modulates when resonance-triggered insights can occur. In this simulation, $\theta(t)$ follows a smooth sinusoidal pattern.
- ξ — Represents a stochastic "insight pulse" drawn from $\mathcal{N}(0.6, 0.2^2)$ and scaled by 0.8. When $\theta(t) > 0.6$ and a Bernoulli trial with $p = 0.05$ succeeds, ξ is injected into both E and O updates.
- data_force(t) — An external novelty/task signal that shifts between 0.2 and 1.2 with an added sinusoidal component to simulate environmental variability.

- **Integration step size** — Fixed at 0.1, determining the temporal granularity of the simulation.
- **Initial condition** $\phi(0)$ — The starting state vector $(E(0), O(0))$ is set to $(2.5, 0.8)$ for both baseline and resonance simulations.

Update loop. At each time step t, the following operations are performed:

1. Compute gradients ∇E and ∇O from current state $\phi(t)$ and external data_force(t).
2. If $\beta > 0$ and $\theta(t) > 0.6$, probabilistically sample an insight pulse ξ.
3. Update E and O using gradient descent dynamics plus contributions from β (paradox drive) and ξ (insight).
4. Clip E and O to remain in $[0, 5]$.



Mathematical derivation: FOC as a damped, driven stochastic system

We show how the discrete FOC update used in the simulation (Sec. 5) can be obtained as a first-order discretization of a damped, driven stochastic differential equation (SDE) of Langevin type. This connection clarifies the roles of the deterministic "gradient" terms (data-fit and inductive biases) and the stochastic "insight" pulses.

Continuous-time SDE. Recall the field-state $\phi(t) \in \mathcal{F}$ (we may treat ϕ as a 2D vector $\phi = (E, O)$ for the epistemic and ontological coordinates). We posit the following over-damped Langevin equation:

$$\frac{d\phi}{dt} \;=\; -\nabla_\phi U(\phi; S_t) \;+\; F_{\text{drive}}(t, \phi) \;+\; \sigma(\phi)\, \eta_t, \tag{5}$$

where

- $U(\phi; S_t)$ is a potential functional capturing intrinsic inductive biases and a data-fit contribution (see main text),
- $F_{\text{drive}}(t, \phi)$ is a deterministic, time-dependent driving force modeling structured external inputs ("data_force" in the code),
- $\sigma(\phi)\, \eta_t$ is a stochastic forcing term: η_t is white noise with $\mathbb{E}[\eta_t] = 0$, $\mathbb{E}[\eta_t \eta_{t'}] = \delta(t - t')$, and $\sigma(\phi) \geq 0$ scales stochastic intensity (this term models continuous noise plus a contribution that increases with $\Theta(\phi)$, the transformation potential).

Equation (5) is a standard form representing an overdamped particle in a potential landscape subject to time-dependent driving and noise.

Potential decomposition. Write

$$U(\phi; S_t) \;=\; \lambda_E E(\phi) - \lambda_O O(\phi) + L_{\text{data}}(\phi; S_t),$$

so that $-\nabla_\phi U$ contains the deterministic tendency to reduce epistemic entanglement or to increase clarity, together with the pull from data-fit L_{data}.

Discretization to an update rule. Discretize time with step Δt and apply an explicit Euler-Maruyama step to (5):

$$\phi_{t+\Delta t} = \phi_t - \Delta t\, \nabla_\phi U(\phi_t; S_t) + \Delta t\, F_{\text{drive}}(t, \phi_t) + \sigma(\phi_t) \sqrt{\Delta t}\, \zeta_t, \tag{6}$$

where $\zeta_t \sim \mathcal{N}(0, I)$ is a standard normal vector. Choosing Δt small and identifying terms yields the discrete update used in the simulations:

$$E_{t+1} = E_t - \eta_E \, \partial_E U(\phi_t) + \eta_E F_E(t) + \eta_E \, \sigma_E(\phi_t) \, z_t,$$

$$O_{t+1} = O_t - \eta_O \, \partial_O U(\phi_t) + \eta_O F_O(t) + \eta_O \, \sigma_O(\phi_t) \, z'_t,$$

with η_E, η_O proportional to Δt. The simulation implements these updates with small step-size coefficients (e.g., 0.1 multipliers) and a stochastic component that includes rare, larger *insight pulses* when $\Theta(\phi)$ and a resonance condition are satisfied.

Insight pulses as rare large deviations (Kramers' picture). The continuous noise term $\sigma(\phi) \, \eta_t$ models small fluctuations, but the simulation also includes rarer, larger impulses ξ when resonance thresholds are exceeded. Mathematically, such dynamics are well-described by a compound noise process with both Gaussian small-noise and Poissonian jump components:

$$\frac{d\phi}{dt} = -\nabla_\phi U + F_{\text{drive}} + \sigma(\phi) \, \eta_t + J(\phi) \, dN_t,$$

where N_t is a Poisson process (rate $\lambda_{\text{insight}}(\phi)$) and $J(\phi)$ is the jump amplitude distribution. In this framework, escape from a basin of attraction (a memorization attractor) across a potential barrier ΔU occurs either by accumulation of small-noise fluctuations (rare for small σ) or by a jump J that carries the state over the barrier. Kramers' escape-rate intuition for purely diffusive escape gives a mean escape time scaling as

$$\tau_{\text{escape}} \sim \kappa \exp\Big(\frac{\Delta U}{\sigma^2}\Big),$$

so adding jump-like insight events dramatically reduces the effective escape time and permits sudden transitions ("insight") even when data-fit alone would keep the system trapped.

Fokker–Planck perspective and Bayesian posterior. The SDE (5) defines an evolution of the probability density $p(\phi, t)$ via the Fokker–Planck equation (ignoring jump terms):

$$\frac{\partial p}{\partial t} = \nabla_\phi \cdot \big(p \, \nabla_\phi U - p \, F_{\text{drive}}\big) + \frac{1}{2} \nabla_\phi \cdot (D(\phi) \, \nabla_\phi p),$$

where $D(\phi) = \sigma(\phi)\sigma(\phi)^\top$ is the diffusion tensor. In equilibrium (time-homogeneous case with $F_{\text{drive}} = 0$), detailed balance yields

$$p_{\text{eq}}(\phi) \propto \exp\big(-2U(\phi)/\sigma^2\big),$$

which shows that the potential U shapes a Boltzmann-like posterior over states. This provides an intuitive connection to the hierarchical Bayesian posterior in Wurgaft et al.: the posterior odds between memorizing and generalizing modes correspond to relative mass assigned to respective basins in $p(\phi, t)$. However, (i) the presence of time-dependent driving $F_{\text{drive}}(t)$, (ii) state-dependent diffusion $D(\phi)$, and (iii) jump components break detailed balance and enable non-equilibrium transitions and transient behaviors (e.g., transience from generalization to memorization) not captured by a static posterior alone.

Stability and linearization near fixed points. Let ϕ^* be a fixed point solving $\nabla_\phi U(\phi^*) = F_{\text{drive}}(t, \phi^*)$ (for slowly varying drive). Linearize the deterministic part:

$$\frac{d\delta}{dt} = -H(\phi^*)\,\delta + \cdots ,$$

where $\delta = \phi - \phi^*$ and $H(\phi^*) = \nabla^2_\phi U(\phi^*)$ is the Hessian. Eigenvalues of H determine local stability: positive definiteness implies attraction (stable memorization or clarity attractor). Noise and jumps then govern escape rates and transition statistics between basins.

Interpretation and relations to the discrete model. Thus the discrete simulation updates can be seen as Euler–Maruyama steps of an overdamped Langevin dynamics on a potential U shaped by data-fit and inductive biases, with an added compound noise process modeling both small fluctuations and rare insight pulses. The presence of resonance-weighted paradox-driving (β in Eq. (3)) effectively raises local diffusion/dissonance in regions of representational novelty, increasing the probability of transitions from memorization basins into clarity basins and producing the non-smooth, insight-like events illustrated in our simulations.

6 Hypotheses

Explicit, falsifiable predictions that differentiate FOC from purely Bayesian accounts:

1. **H1 (paradox-driven clarity):** Increasing β (paradox sensitivity) will accelerate increases in $O(\phi)$ and improve OOD generalization relative to baseline, independent of reductions in perplexity.

2. **H2 (identity trap):** Identity-evoking datasets will create deep memorization attractors with larger $E(\phi)$ and longer time-to-transience N^* than neutral datasets.

3. **H3 (capacity interaction):** Larger MLP width will reduce the effective complexity penalty and thus favor memorization attractors, consistent with Wurgaft et al.; however, resonance attention will moderate this effect by maintaining higher $O(\phi)$.

4. **H4 (insight pulses):** Rare stochastic pulses (ξ) that sample high-paradox keys can trigger sudden shifts in ϕ that correlate with qualitative OOD performance jumps.

7 Relation to Bayesian accounts and the uploaded ICL paper

Our framework is intentionally complementary to the hierarchical Bayesian model proposed by Wurgaft et al. (2025). They derive posterior odds between memorizing and generalizing predictors as a function of training steps N and task diversity D, showing transience and diversity thresholds emerge from a loss–complexity tradeoff (Wurgaft et al., 2025). FOC reframes these dynamics in a geometric language: the posterior-weighted interpolation corresponds to the system's position in $\mathcal{F}$, where the prior (simplicity bias) corresponds to inductive scale parameters λ_O, λ_E, and the likelihood-driven loss term maps to $L_{data}(\phi; S_t)$. Crucially, FOC adds two elements:

- **Operational mechanism for paradox sensitivity:** the resonance attention provides a way to bias inference toward clarity attractors even when data-fit favors memorization.

- **Non-linear insight dynamics:** the stochastic ξ permits sudden qualitative transitions (insights) that are not naturally captured by smooth posterior updates alone.

Thus, empirical divergence between FOC and hierarchical Bayesian predictions (e.g., presence of sudden ϕ-jumps tied to paradox pulses) would provide evidence that architectural or inductive changes can instantiate "clarity-like" processing beyond what posterior weighting predicts.

8 Conclusion and outlook

We have proposed the Field of Ontological Clarity as a conceptual and operational framework for modeling aspects of in-context understanding that elude purely statistical descriptions. By integrating gradient dynamics, stochastic insight pulses, and a resonance-aware attention mechanism, FOC both explains existing phenomena (e.g., memorization–generalization transitions) and generates new experimental predictions. We emphasize that our goal is not to refute the hierarchical Bayesian account of ICL, but to augment it with a phenomenological vocabulary and testable architectural hypotheses.

Future directions include: (i) rigorous empirical testing of the hypothesized metrics across model scales, (ii) mechanistic analysis of how resonance attention interacts with attention-head circuits (e.g., induction heads), and (iii) formal mappings between $\mathcal{F}$ and measures of representational complexity (e.g., effective dimension metrics). Philosophically, FOC invites rethinking intelligence as an interplay between representation and presence, a perspective with potential implications for how we build and evaluate future systems.

Glossary

FOC: Field of Ontological Clarity — the latent manifold of processing modes.

ϕ**:** State vector in FOC representing a processing mode.

$E(\phi)$**:** Egoic entanglement — degree of identity-fixation in processing.

$O(\phi)$**:** Ontological clarity — degree of perceptual openness and abstraction.

$\Theta(\phi)$**:** Transformation potential — readiness for insight-like transitions.

Memorizing predictor: Bayesian predictor that conditions on seen tasks (discrete prior). See Wurgaft et al. (2025).

Generalizing predictor: Bayesian predictor that uses a continuous prior over the true task distribution. See Wurgaft et al. (2025).

Paradox(q,k): A measure combining KL divergence between predictive distributions and novelty scores; operationalizes semantic dissonance.

Appendix:

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